

RESEARCH ARTICLE

Association of common and rare variants with Alzheimer's disease in more than 13,000 diverse individuals with whole-genome sequencing from the Alzheimer's Disease Sequencing Project

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Abstract

INTRODUCTION: Alzheimer's disease (AD) is a common disorder of the elderly that is both highly heritable and genetically heterogeneous.

METHODS: We investigated the association of AD with both common variants and aggregates of rare coding and non-coding variants in 13,371 individuals of diverse ancestry with whole genome sequencing (WGS) data.

RESULTS: Pooled-population analyses of all individuals identified genetic variants at apolipoprotein E (APOE) and BIN1 associated with AD ($p < 5 \times 10^{-8}$). Subgroup-specific analyses identified a haplotype on chromosome 14 including PSEN1 associated with AD in Hispanics, further supported by aggregate testing of rare coding and non-coding variants in the region. Common variants in LINC00320 were observed associated with AD in Black individuals ($p = 1.9 \times 10^{-9}$). Finally, we observed rare non-coding variants in the promoter of TOMM40 distinct of APOE in pooled-population analyses ($p = 7.2 \times 10^{-8}$).

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DISCUSSION: We observed that complementary pooled-population and subgroup-specific analyses offered unique insights into the genetic architecture of AD.

KEYWORDS

Alzheimer's disease, genetics, rare genetic variants, whole genome sequencing

Highlights

- We determine the association of genetic variants with Alzheimer's disease (AD) using 13,371 individuals of diverse ancestry with whole genome sequencing (WGS) data.
- We identified genetic variants at apolipoprotein E (*APOE*), *BIN1*, *PSEN1*, and *LINC00320* associated with AD.
- We observed rare non-coding variants in the promoter of *TOMM40* distinct of *APOE*.

1 | BACKGROUND

Alzheimer's disease (AD) is a progressive neurological disorder characterized by a decline in cognitive and memory functions, which ultimately results in the inability to carry out daily activities. It is the most common cause of dementia, affecting > 50 million people worldwide. This number is projected to almost triple by 2050, reaching 152 million people, as the baby boom generation (born between 1946 and 1964) has already begun to reach age 65 years and beyond.¹ The disease is more common among older individuals, with the risk increasing significantly after the age of 65 years. In the United States, it is estimated that $\approx$ 6.5 million people currently have AD, with a projection of nearly 14 million by 2050.¹

Although the underlying multidimensional causes of AD are not fully understood, evidence suggests that genetics plays a crucial role in the development of the disease. Rare coding changes in presenilin 1 (*PSEN1*),² presenilin 2 (*PSEN2*),³ and amyloid precursor protein (*APP*)^{4–7} underlie autosomal dominant early-onset AD, while other coding changes in these genes are associated with an increased risk of late-onset AD. The apolipoprotein E (*APOE*) gene is the strongest susceptibility gene associated with AD,^{8,9} with isoforms defined by common missense variants associated with large effects on AD risk. Individuals with one copy of the *APOE* ϵ 4 allele have approximately a 3-fold increased risk of developing AD, while those with two copies of the ϵ 4 allele are at approximately a 12-fold increased risk.¹⁰ The presence of the *APOE* ϵ 4 allele is also associated with an earlier onset of AD.¹¹ Each of these loci was first identified by family-based studies.

Recent large-scale genome-wide association studies using array-based genotyping and imputation have identified > 80 common genetic loci associated with AD.^{12–14} It is estimated that 25% of phenotypic variation in AD remains unexplained by known genetic variants associated with AD,^{15,16} suggesting that additional risk loci are yet to be discovered. While genotype arrays are a useful tool for studying genetic variants associated with AD, they have limitations when it comes to discovering rare or novel genetic variant associations with

disease. Array-based genotyping relies on a predesigned set of probes that target specific genetic loci across the genome. Computational imputation may mitigate this limitation and improve the accuracy and coverage of array-based genotype data. However, imputation accuracy is directly determined by the size and quality of the reference panel and observed array data used, as well as the underlying patterns of genetic variation in the populations being studied. In contrast, whole genome sequencing (WGS) enables a full-spectrum exploration of short insertion/deletions (INDELs) and single nucleotide variants (SNVs) across the genome and provides a comprehensive view of an individual's genetic information, allowing for the testing of both common and rare genetic variants that may be unique to individuals or populations not previously observed.

The Alzheimer's Disease Sequencing Project (ADSP) is a collaborative research effort that uses WGS to identify both protective and risk genetic factors for AD. Data are collected from diverse individuals, recognizing that genetic risk, incidence, and prevalence rates can vary across populations. For example, *APOE* ϵ 4 is more common in White individuals ($\approx$ 24%) and Black individuals ($\approx$ 26%) compared to Asian individuals ($\approx$ 12%) or Hispanic individuals ($\approx$ 15%) and the effect of *APOE* ϵ 4 varies among populations.^{17,18} The odds ratios of ϵ 3/ ϵ 4 are 2.49 and 3.83 for African ancestry and European ancestry, respectively, while ϵ 4/ ϵ 4 are 8.17 and 14.35, respectively.¹⁹ Leveraging large-scale WGS from the ADSP, we performed association testing of single variants (minor allele frequency [MAF] > 0.5%) as well as aggregates of rare (MAF < 1%) coding and non-coding variants in up to 13,371 individuals ($N_{\text{cases}} = 6519$ and $N_{\text{control}} = 6852$).

2 | METHODS

2.1 | Study participants

Data from the ADSP are available to qualified investigators via the National Institute on Aging Genetics of Alzheimer's Disease Data

RESEARCH IN CONTEXT

1. **Systematic review:** We examined peer-reviewed (PubMed) and preprint articles using whole genome sequencing (WGS) data to determine genetic variants associated with Alzheimer's disease (AD). We performed association analyses using Alzheimer's Disease Sequencing Project WGS data to determine common variants and aggregates of rare variants with AD.
2. **Interpretation:** We observed genetic variants at *APOE*, *BIN1*, *PSEN1*, and *LINC00320* associated with AD. Furthermore, we observed an aggregate of rare coding variants in *PSEN1* and aggregated rare non-coding variants in the promoter of *TOMM40* associated with AD.
3. **Future directions:** Our study provides insights into rare variations associated with AD. Future studies will be needed to functionally validate these variants.

Storage Site (NIAGADS: <https://dss.niagads.org/>). The current analyses focused on participants with WGS in the NIAGADS data named "R3 17K WGS Project Level VCF." WGS data have been generated in multiple cohorts as part of the ADSP. Samples were sequenced by multiple centers with different platforms and libraries (Table S1, [Supplementary Methods](#) in supporting information). The Genome Center for Alzheimer's Disease (GCAD) mapped short reads against the reference genome hg38 using BWA MEM,²⁰ called variants using the Genome Analysis Toolkit (GATK)²¹ HaplotypeCaller for each sample, and then performed joint genotyping across all samples using GATK.²² The GCAD quality control (QC) working group performed quality checks of variants and genotypes and assigned a quality annotation.²³ Details of studies included in the ADSP can be found at NIAGADS under dataset: NG00067 ADSP Umbrella Study (<https://dss.niagads.org/datasets/ng00067/>). Of the 16,905 individuals in the ADSP Release 3 data, individuals with unknown AD status and genetically identical individuals were excluded using GCAD recommendations for excluding genetically identical individuals. After removing these individuals, 13,371 individuals were available for association analysis with AD.

We created three groups for subgroup-specific analyses. Individuals who reported their race as White and their ethnicity as non-Hispanic or missing were classified as non-Hispanic White (NHW). Individuals who indicated any reported race and Hispanic ethnicity were classified as Hispanic (HIS). Individuals who reported their ethnicity as non-Hispanic or had missing ethnicity and their race as Black were classified as African American (AA). There were 179 individuals that were not classified into one of our subgroup-specific analyses. Using genetic similarity clustering did not substantially change the subgroups. The results of subgroup-specific analyses were also confirmed by groups clustered using genetic information.

2.2 | Sample clustering using genetic similarity

To cluster samples based on genetic information, we used the following approach. First, we calculated the Euclidean distance of PC1-2 between each sample and the three reference populations from the 1000 Genomes Project,^{24,25} Europeans (EUR), Africans (AFR), and East Asians (EAS). Then, we used a second approach to create sample clusters, which involved conducting global ancestry inference analysis with GRAF-pop,²⁶ which uses 100,437 fingerprint single nucleotide polymorphisms (SNPs) and is a principal component analysis (PCA)-free method for ancestry inference. Based on the results of GRAF-pop, we clustered the samples into subgroups: African-, Latin-, and European-ancestry groups.

2.3 | AD phenotype definition

The ADSP provides different AD status variable definitions for participants included via case-control versus family-based studies. Additionally, distinct phenotype data are available for some augmentation studies. In the current analysis, for individuals in the ADSP case-control study, we defined AD cases as individuals with either prevalent or incident AD. Case-control individuals with no prevalent or incident AD were defined as controls and those with NA for status were defined as unknown. In the ADSP family phenotype file, possible values for the AD status variable include no dementia, definite AD, probable AD, possible AD, family-reported AD, other dementia, family reported no dementia, and unknown. For family-based individuals, we defined an AD case as either possible, probable, or definite AD. AD controls were defined as individuals coded as no dementia. We redefined individuals with family-reported AD, other dementia, or unknown status as missing AD status. The Alzheimer's Disease Neuroimaging Initiative phenotype data, which is part of the ADSP augmentation study, provides information on mild cognitive impairment (MCI) in addition to AD status. Individuals with a current diagnosis of MCI were excluded in the current study. Each study contributing data to ADSP (see [supporting information](#)) adjudicates their participants independently.

2.4 | Genotype calling and quality control

The data used in this study were processed by GCAD. GCAD processed all WGS samples received in FASTQ, BAM, or compressed BAM (CRAM) formats using the official GCAD/ADSP pipeline, VCPA1.1.²² In brief, GCAD aligned the samples against GRCh38/hg38 with BWA-mem and called variants, including SNVs and short indels, using GATK 4.1.1. QC was performed on all samples and variants following the GCAD/ADSP QC pipeline.²³ Various filtering and quality-checking strategies were applied at both the genotype and variant levels. Each genotype was set to missing ("./.") if either the read depth (DP) was < 10 or the genotype quality (GQ) score was < 20. Variants of low quality were flagged but not excluded from the datasets. For variants, flags were applied in the following order: (1) variants in GATK low

sequence quality tranches (variants without a FILTER value of "PASS" that are above the 99.8% VQSR Tranche), (2) monomorphic variants were flagged, (3) variants with high missing rate were flagged, and (4) variants with high DP were flagged. ABHet is a variant-level annotation estimating if biallelic variants match expected allelic ratios, with an ideal heterozygous variant having a value close to 0.5. Variants used for analysis were those without QC flags, with ABHet in the range of 0.25 to 0.75, and with a missing rate of < 5%.

2.5 | Principal component analysis

PCA is widely used for analyzing large datasets that have a high number of dimensions or features per observation. PCA is a statistical technique for reducing the dimensionality of a dataset, while still retaining as much of the original variation as possible. In genetic studies, PCA is commonly used to infer population structure in the data, because population structure is a major factor that affects sample genotypes. Typically, the top principal components (PCs) calculated from the genotype data reflect population structure among the individuals. To ensure accurate ancestry inference, we used PC-AiR in the GENESIS²⁷ package for a PCA, a tool that accounts for sample relatedness and thus provides accurate ancestry inference. We calculated PCs for the pooled samples using variants with MAF > 5%, call rate > 99%, GCAD provide variant flag (VFlag) = 0 (no exclusion),²³ RUTH²⁸ Hardy-Weinberg Equilibrium (HWE) p value > 10^{-4} , and excluded variants in high linkage disequilibrium (LD) regions (https://genome.sph.umich.edu/wiki/Regions_of_high_linkage_disequilibrium_LD) and variants in LD using an $r^2 < 0.1$. For the AA, HIS, and NHW subgroups, we calculated PCs using SNVs with MAF > 5%, VFlag = 0, and LD threshold r^2 (0.1).

To identify the locations of *PSEN1* p.G206A (rs63750082) carriers, we extended the PC calculation to include samples of the 1000 Genomes Project. The VCFs of the 1000 Genomes Project were downloaded and merged with ADSP R3 VCFs by bcftools. Population and super-population information of each sample was also downloaded from the 1000 Genomes Project. The MAF cutoff, >5%, was applied, and then PC calculation was performed.

2.6 | Covariates, analysis models, and single variant analysis

We included sex, technical sequencing variables (sequencing center, sequencing length, and polymerase chain reaction [PCR] status), and PCs (PC1–5 for subgroup-specific analysis and PCs associated with AD status for the pooled samples analysis) as covariates in all our models along with a genetic relationship matrix to adjust for relatedness among individuals. As a secondary model, we additionally adjusted for the number (0, 1, or 2) of *APOE* ϵ 4 alleles and the number of *APOE* ϵ 2 alleles. We tested each variant with a MAF > 0.5% for association with AD using the score test in the GENESIS²⁹ package to fit a penalized quasi-likelihood (PQL) approximation to the generalized linear

mixed model. Variants that failed HWE by RUTH²⁸ ($\text{HWE_SLP_I} < -4$ or $\text{HWE_SLP_I} > 4$) in controls were excluded as well as variants in low complexity regions. We used a standard significance threshold of 5×10^{-8} for our single variant association analyses.

2.7 | Aggregates of rare variants analysis

To test aggregates of coding and non-coding rare variants, we implemented the STAAR pipeline³⁰ using both SNVs and INDELs. The STAAR pipeline is a set of routines for performing association analysis of large-scale WGS data using the STAAR framework³¹ to aggregate rare variants using variant set analysis for both gene-centric coding and gene-centric non-coding analysis. We used the STAAR- O p value, which combines p values across multiple annotation-weighted variant set tests.³¹

The gene-centric coding analysis of the STAAR pipeline provides five genetic categories: putative loss of function (pLoF), missense, disruptive missense, pLoF and disruptive missense, and synonymous. The gene-centric non-coding analysis provides eight genetic categories: promoter or enhancer overlaid with Cap Analysis of Gene Expression (CAGE) or DNase Hypersensitivity (DHS) sites, untranslated region (UTR), upstream, downstream, and non-coding RNA genes. We set our significance threshold for our rare variant aggregation tests to be 1×10^{-7} (Bonferroni correction for testing $\approx 20,000$ genes across five coding categories and eight non-coding categories). For gene-centric non-coding analysis, due to the known associations in *PSEN1* and *APOE* regions, we performed conditional analyses adjusting for *PSEN1* p.G206A (rs63750082) in chromosome 14 and *APOE* ϵ 2 and ϵ 4 alleles in chromosome 19 in the pooled samples analysis. To incorporate additional features of the STAAR pipeline, we created a GitHub repository that performs variant extraction and conditional analysis (see code availability).

2.8 | Global and local ancestry inference analysis

The global ancestry inference was performed using the GRAF-pop²⁶ tool, which uses 100,437 fingerprint SNPs and is a PCA-free method for ancestry inference. GRAF-pop provides results of comparable quality to PCA-based methods such as EIGENSTRAT, FastPCA, and FlashPCA2, while offering an ultra-fast running time. Genotypes were provided to GRAF-pop in VCF format. The tool assumes that each individual is a mixture of three ancestries: European (E), African (F), and Asian (A), and estimates the ancestral proportions P_e , P_f , and P_a using barycentric coordinates.

To infer local ancestry, specifically for the analysis in the 14q24 region, we used RFMIX³² with 2504 reference genomes from the 1000 Genomes Project. This tool outperforms other methods for estimating local ancestry in complex admixture scenarios.³³ Prior to using RFMIX, we phased variants using SHAPEIT4.³⁴ In addition, we used PICARD (<http://broadinstitute.github.io/picard/>) to LiftOver coordinates from hg38 to hg19 as the genetic map of reference samples

we used was against HG19 (https://mathgen.stats.ox.ac.uk/impute/1000GP_Phase3.html). RFMIX allows for a comprehensive understanding of the local ancestry composition in the specific region of interest and provides insights into the complex genetic makeup of diverse populations.

2.9 | INDEL experimental validation

PCR Primer Design: The genomic sequence for the INDEL variants was determined by submitting the chromosomal location of the variants to the December 2013 (GRCh38/hg38) version of the Genome Browser³⁵ (<http://genome.ucsc.edu>). Sequence surrounding the variants was extracted and used to design PCR primers. Primers were designed outside of the breakpoints to amplify across the insertion/deletion sequence. All PCR primer sequences were submitted to the Blast-like alignment tool (BLAT) to check for the uniqueness of the sequences. When available, samples from three individuals reported as homozygous or heterozygous for the variant were used for sequence validation along with one control (or reference) sample. When possible, samples from multiple families were used for validation.

PCR and Sanger Sequencing: Genomic DNA (≈ 50 ng) was amplified using a SimpliAmp Thermal Cycler (Applied Biosystems) in a 20 μ L reaction volume with HotStarTaq Master Mix (Qiagen) in the presence of 2 μ M primers (IDT). The PCR conditions used were: 95°C 15 minutes followed by 30 cycles of 95°C 20 seconds, 55°C 30 seconds, 72°C 2 minutes with a final extension of 72°C 7 minutes. The amplified PCR products were prepared for Sanger sequencing by adding ExoSAP-IT (USB) and incubating at 37°C for 45 minutes followed by 80°C for 15 minutes. The PCR products were then Sanger sequenced using the BigDye Terminator v3.1 Cycle Sequencing kit (Part No. 4336917, Applied Biosystems). The sequencing reaction contained BigDye Terminator v3.1 Ready Reaction Mix, 5X Sequencing Buffer, 5 M Betaine solution (Part No. B0300 Sigma), and 0.64 μ M sequencing primer (IDT) in a total volume of 5 μ L. The sequencing reaction was performed in a SimpliAmp Thermal Cycler (Applied Biosystems) using the following program: 96°C 1 minute followed by 25 cycles of 96°C 10 seconds, 50°C 5 seconds, 60°C 1 minute 15 seconds. The products were cleaned using XTerminator and SAM Solution (Applied Biosystems) with 30 minutes of shaking at 1800 rpm (36 g) followed by centrifugation at 1000 rpm (200g) for 2 minutes. The sequencing products were analyzed on a SeqStudio Genetic Analyzer (Applied Biosystems) and the sequencing traces were analyzed using Sequencher 5.4 (Gene Code).

2.10 | INDEL in silico validation

Based on our findings in this study, we encountered 12 significant INDELs that were experimentally validated as false positives. These INDELs were all situated within sequences that exhibited discrepancies between the Telomere-to-Telomere Consortium (T2T)³⁶ and GRCh38 (the reference genome used in our study). Consequently,

INDELs located in these discrepant sequences between T2T and GRCh38 were excluded.

To address this issue, we initially identified regions by considering the coordinates of the INDELs along with their respective lengths and then extended these regions by ± 10 bp. Subsequently, we used LiftOver, using the hg38-to-chm13v2 chain (available at <https://hgdownload.soe.ucsc.edu/goldenPath/hs1/liftOver/hg38-chm13v2.over.chain.gz>) to convert the regions to the corresponding coordinates in the T2T (chm13v2) assembly. In cases in which the LiftOver process was unsuccessful, the INDELs were excluded from further analysis. For the regions that successfully underwent LiftOver, we compared the sequences obtained from GRCh38 and T2T. Only when the sequences were found to be identical, we included the corresponding INDELs in our report, ensuring accuracy and reliability in our findings.

2.11 | Replication dataset

The replication dataset was formed based on ADSP R4 36k samples by: (1) calculating identity-by-descent (IBD) for each pair of samples, (2) filtering samples with IBD > 0.4, and (3) excluding samples from R3 17k. This resulted in 17,266 samples (4696 cases and 9782 controls; Table S2 in supporting information). The dataset includes 2176 Black individuals (AA, $N_{\text{cases}} = 307$, $N_{\text{controls}} = 1366$, $N_{\text{unknown}} = 503$), 7452 Hispanics (HIS, $N_{\text{cases}} = 1879$, $N_{\text{controls}} = 4343$, $N_{\text{unknown}} = 1230$), and 4879 non-Hispanic Whites (NHW, $N_{\text{cases}} = 2243$, $N_{\text{controls}} = 1617$, $N_{\text{unknown}} = 1019$).

3 | RESULTS

3.1 | Overview

We performed association analyses using 13,371 of the 16,905 individuals in the ADSP Release 3 data to discover common and rare genetic variants associated with AD. The individuals who were excluded from the analysis represent technical replicates, unexpected duplicates, and individuals with unknown AD status. In the 13,371 individuals, 177.6 million bi-allelic SNVs and 12.9 million bi-allelic INDELs were observed.

Given the ADSP data are composed of diverse individuals, we performed association testing across all participants (pooled samples, $N_{\text{cases}} = 6519$ and $N_{\text{control}} = 6852$) and within the three subgroups: AA ($N_{\text{cases}} = 1137$ and $N_{\text{control}} = 1707$), HIS ($N_{\text{cases}} = 1021$ and $N_{\text{control}} = 1988$), and NHW ($N_{\text{cases}} = 4230$ and $N_{\text{control}} = 3109$) defined by reported race and ethnicity (Figure 1; Table 1). We performed single variant association testing (MAF > 0.5%) as well as association testing of aggregates of rare (MAF < 1%) coding and non-coding variants within the pooled samples and each of the three subgroups (AA, HIS, and NHW). The pooled samples analysis is most powerful to detect associations when there are similar frequencies and effect sizes across subgroups while the subgroup-specific analyses are beneficial to detect subgroup-specific effects. We limited the analyses to bi-allelic

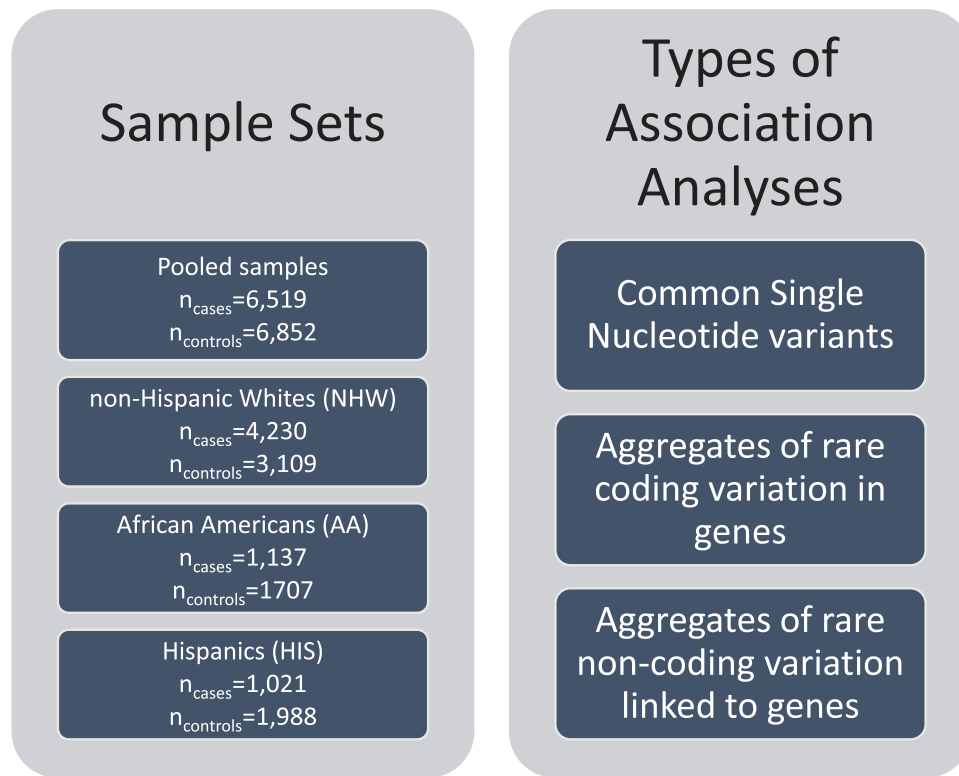


FIGURE 1 Study overview. Three types of association analyses in four sets of individuals were performed, pooled samples, non-Hispanic Whites (NHW), African Americans (AA), and Hispanics (HIS). The pooled samples set included all individuals in the NHW, AA, and HIS sets, plus individuals that were not defined to be in one of those subsets.

TABLE 1 Descriptive statistics of ADSP study samples used for analyses.

	AD				No AD			
	PS ^b (N = 6519; 100%)	AA (N = 1137; 17.4%)	Hispanic (N = 1021; 15.7%)	NHW (N = 4230; 64.9%)	PS ^c (N = 6852; 100%)	AA (N = 1707; 24.9%)	Hispanic (N = 1988; 29.0%)	NHW (N = 3109; 45.4%)
Sex								
Female	3921 (60.1%)	798 (70.2%)	654 (64.1%)	2379 (56.2%)	4580 (66.8%)	1293 (75.8%)	1374 (69.1%)	1886 (60.7%)
Age								
Mean (SD)	71.6 (18.1)	74.6 (10.1)	74.4 (13.6)	70.1 (20.5)	74.5 (16.6)	75.1 (8.8)	65.1 (26.0)	80.2 (6.7)
APOE ε4 alleles ^a								
0	3022 (46.4%)	424 (37.3%)	677 (66.3%)	1876 (44.3%)	4608 (67.3%)	1112 (65.1%)	1550 (78.0%)	2005 (64.5%)
1	2917 (44.7%)	547 (48.1%)	288 (28.2%)	2016 (47.7%)	1993 (29.1%)	555 (32.5%)	415 (20.9%)	1004 (32.3%)
2	570 (8.7%)	166 (14.6%)	56 (5.5%)	338 (8.0%)	164 (2.4%)	40 (2.3%)	23 (1.2%)	100 (3.2%)
APOE ε2 alleles ^a								
0	6068 (93.1%)	995 (87.5%)	930 (91.1%)	4027 (95.2%)	5930 (86.5%)	1309 (76.7%)	1788 (89.9%)	2877 (92.5%)
1	427 (6.6%)	136 (12.0%)	89 (8.7%)	198 (4.7%)	800 (11.7%)	375 (22.0%)	196 (9.9%)	224 (7.2%)
2	14 (0.2%)	6 (0.5%)	2 (0.2%)	5 (0.1%)	35 (0.5%)	23 (1.3%)	4 (0.2%)	8 (0.3%)

Abbreviations: AA, Black African American; AD, Alzheimer's disease; APOE, apolipoprotein E; NHW, non-Hispanic White; PS, pooled samples; SD, standard deviation; WGS, whole genome sequencing.

^aε2 and ε4 are derived from APOE genotype not WGS.

^b131 (2.0%) samples are reported other ethnicities.

^c48 (0.7%) samples are reported other ethnicities.

SNVs and INDELs after preliminary analyses showed false-positive associations across the genome for multi-allelic variants.

Sample clustering by using genetic similarity, based on the Euclidean distance of PC1-2 between each sample and the reference populations from the 1000 Genomes Project, indicated that 92.94% and 7.06% of reported AA samples were found to be closer to the AFR and EUR reference populations, respectively (Figure S1 in supporting information). Additionally, 8.28%, 0.003%, and 91.69% of reported HIS samples were closer to the AFR, EAS, and EUR reference populations, respectively. Similarly, 0.08%, 0.01%, and 99.90% of reported NHW samples were closer to the AFR, EAS, and EUR reference populations, respectively. This result aligns with the analysis from GRAF-pop, which revealed that 97.78% of reported AA samples were inferred to have a predominantly African ancestry, while 99.43% of reported NHW samples were inferred to have a predominantly European ancestry (Figure S2, Table S3 in supporting information). Regarding reported HIS samples, 74.74% were inferred to have Latin American 1 ancestry, 20.51% had African American ancestry, 2.43% had Latin American 2 ancestry, and 1.46% had European ancestry. Based on the results of GRAF-pop, we clustered the samples African and African American as African-ancestry group ($N = 3453$), Latin American 1 and Latin American 2 as Latin-ancestry group ($N = 2403$), and European as European-ancestry group ($N = 7440$). Subsequently, we conducted association analyses within these genetically defined groups. Nevertheless, the findings from these analyses revealed no notable differences compared to the previous subgroup analyses based on self-reported ethnicity (Figure S3 in supporting information).

3.2 | Association of single variants with AD

To determine individual SNVs and INDELs associated with AD status, we performed individual variant association testing. As expected, we found the strongest associations at the *APOE* locus (chr19:44,905,796-44,909,393), the major genetic risk factor for AD.^{8,37} These associations were observed in the pooled samples analysis as well as the analysis within each of the three subgroups (Figures 2 and S4 in supporting information). We observed that the $\epsilon 4$ haplotype (the alternative allele at rs429358 and reference allele at rs7412) is more common in AD cases as well as most frequent in AA individuals and least frequent in HIS individuals (Table S4 in supporting information). While the odds of AD were higher in the AA individuals, the 95% confidence interval for the AA individuals overlaps with the confidence interval for the NHW individuals (Figure 2). We observed the $\epsilon 2$ haplotype (the reference allele at rs429358 and alternative allele at rs7412) was enriched in controls and had a lower frequency (frequency = 0.05) than the $\epsilon 4$ haplotype (frequency = 0.24). The $\epsilon 2$ haplotype is most frequent in AA individuals, followed by HIS individuals, and least frequent in NHW individuals. We confirmed the association with the *APOE* locus with the replication data (Table S5 in supporting information, pooled samples $p = 3.4 \times 10^{-148}$, AA $p = 2.2 \times 10^{-13}$, HIS $p = 1.7 \times 10^{-36}$, and NHW $p = 9.5 \times 10^{-85}$).

We also observed a genome-wide (GW) significant ($p < 5 \times 10^{-8}$) association of *BIN1* (rs4663105, MAF = 0.47, odds ratio [OR] = 1.17, $p = 3.2 \times 10^{-9}$) with AD status in the pooled samples analysis. Previous studies have identified *BIN1* as an AD susceptibility gene after *APOE*.^{12,14,38} After adjusting for *APOE* $\epsilon 4$ (rs429358) and $\epsilon 2$ (rs7412) alleles on chromosome 19, the association of *BIN1* on chromosome 2 remained largely the same (OR = 1.17, $p = 3.1 \times 10^{-9}$). *BIN1* variants were also associated with AD in the NHW subgroup (rs4663105, MAF = 0.43, OR = 1.22, $p = 1.2 \times 10^{-8}$, Tables 2 and S6 in supporting information). Although the association did not reach statistical significance, the *BIN1* variant showed a consistent direction of effect in the AA and HIS subgroups (AA: rs4663105, MAF = 0.41, OR = 1.16, $p = 0.0064$; HIS: rs4663105, MAF = 0.40, OR = 1.10, $p = 0.15$). We replicated the *BIN1* association in pooled samples and NHW samples (pooled samples $p = 8.3 \times 10^{-03}$, NHW: $p = 2.2 \times 10^{-02}$).

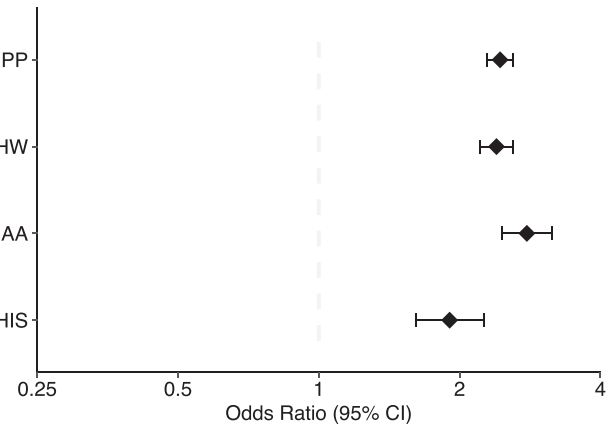
In AA individuals, we observed an association between variants in *LINC00320* with AD (rs144204759, MAF = 0.018, OR = 3.4, $p = 1.9 \times 10^{-8}$; Tables 2 and S7, and Figure S5A in supporting information). This variant has a much lower frequency in NHW individuals (gnomAD frequency = 0.005) and *LINC00320* was previously implicated in AD³⁹ by a distinct variant in the International Genomics of Alzheimer's Project (IGAP) genome-wide association study (GWAS) of 25,170 AD cases and 41,052 cognitively normal controls. We determined the gene expression change using Agora (<https://agora.adknowledgeportal.org/genes/ENSG00000224924/evidence/rna>).

Although it is not among the top genes, *LINC00320* appears to be differentially expressed in AD ($p = 5.8 \times 10^{-03}$ for the dorsolateral prefrontal cortex [DLPFC], $p = 1.1 \times 10^{-02}$ for the posterior cingulate cortex [PCC], and $p = 3.5 \times 10^{-05}$ for the parahippocampal cortex [PHC]). However, our genetic association was not replicated, which may be due to differences in AA sample structure or sample size between the discovery ($N_{\text{case}} = 1137$, $N_{\text{control}} = 1707$) and replication ($N_{\text{case}} = 307$, $N_{\text{control}} = 1366$) data. Additionally, three INDELs near *APOE* are associated with AD in the AA analysis (rs142042446, MAF = 0.040, OR = 2.27, $p = 5.7 \times 10^{-9}$; rs542555887, MAF = 0.12, OR = 1.77, $p = 2.0 \times 10^{-10}$; rs113492558, MAF = 0.07, OR = 2.56, $p = 1.7 \times 10^{-19}$); however, the reduction of these signals after conditioning *APOE* $\epsilon 2$ and $\epsilon 4$ alleles suggests that the impact of these INDELs is not distinct of these common *APOE* haplotypes.

We determined the association of all loci from a recent GWAS of AD¹² within our data. We observed that among of 143 SNVs reported, 75 have a $p < 0.05$ in our dataset (Table S8 in supporting information). However, spurious variants emerged from the association analysis, underlining the need for careful scrutiny of WGS data (Supplementary Methods, Table S9, Figure S6 in supporting information). Several SNVs in *ANK3* were identified as associated with AD but later proven to be false positives supported by low-quality sequencing reads. Similarly, numerous INDELs identified were not confirmed upon further scrutiny, particularly due to biases introduced by using a reference genome that may not accurately represent human genomic diversity. This underscores the necessity for rigorous verification and analysis methods to prevent misinterpretation of WGS data due to false positives.

rs429358 (APOE e4 allele)

Group	MAF	OR	UCL	LCL	P-value
Pooled Samples	0.235	2.441	2.292	2.599	1.98e-170
NHW	0.258	2.398	2.208	2.605	1.04e-95
AA	0.267	2.785	2.468	3.144	6.67e-62
HIS	0.146	1.904	1.610	2.251	4.93e-14



rs7412 (APOE e2 allele)

Group	MAF	OR	UCL	LCL	P-value
Pooled Samples	0.051	0.546	0.484	0.616	6.56e-23
NHW	0.031	0.523	0.427	0.641	3.76e-10
AA	0.099	0.519	0.434	0.622	9.57e-13
HIS	0.052	0.767	0.584	1.008	0.0572

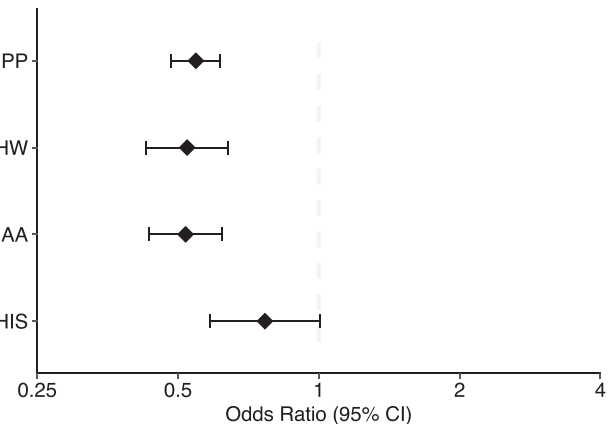


FIGURE 2 Association of APOE alleles with AD by subgroup. AA, African American; AD, Alzheimer's disease; APOE, apolipoprotein E; CI, confidence interval; HIS, Hispanic; LCL, lower confidence limit; MAF, minor allele frequency; NHW, non-Hispanic White; OR, odds ratio; PP, pooled population (samples); UCL, upper confidence limit.

TABLE 2 Genome-wide ($P < 5 \times 10^{-8}$) significant loci associated with AD.

Variants ^a	Gene	RSID	Group	MAF ^b	Odds ratio	pvalue ^c
2-127133851-A-C	<i>BIN1</i>	rs4663105	Pooled samples	0.470	1.16	3.2×10^{-9}
			NHW	0.427	1.22	1.2×10^{-8}
14-73615125-C-T	(various)	rs9671262	HIS	0.005	19.20	2.2×10^{-11}
19-44908684-T-C	<i>APOE</i>	rs429358	Pooled samples	0.230	2.44	2.0×10^{-170}
			AA	0.267	2.78	6.7×10^{-62}
			HIS	0.145	6.69	4.9×10^{-14}
			NHW	0.258	2.40	1.0×10^{-95}
21-20730315-G-A	<i>LINC00320</i>	rs144204759	AA	0.018	3.40	1.9×10^{-9}

Abbreviations: AA, Black or African American; APOE, apolipoprotein E; HIS, Hispanic; MAF, minor allele frequency; NHW, non-Hispanic White.

^aCoordinates in GRCh38 for indexed variant.

^bMAF for subgroup.

^cWhere more than one was significant for a linked gene, the most significant p value, either with or without APOE adjustment, is reported for each gene. The full lists are given in Tables S3–S5 in supporting information.

3.3 | 14q24 in Hispanics

From our single variant association testing, we observed a region from 14q24.2 to 14q24.3 (chr14:72,600,928-75,846,454), with 44

low-frequency variants (43 SNVs and one INDEL; MAF: 0.005–0.012; Tables 2 and S10 in supporting information) associated with AD across 13 genes in Hispanic individuals (Figures S4E, S4F, and S5B). These variants are not associated with AD in the AA subgroup despite a

notably higher allele frequency (MAF: 0.007–0.041; Table S11 in supporting information). Single variant analyses were not conducted for these variants in the pooled samples and NHW subgroups because the MAF in these subgroups was below the 0.5% threshold (Table S11). We observed variants in 14q24.3 replicated ($p < 0.05$) in the Hispanic subgroup ($p = 2.4 \times 10^{-04}$).

This region contains *PSEN1* p.G206A (rs63750082 at 14:73192712), a known early-onset AD causal mutation.^{40–43} After conditioning on *PSEN1* p.G206A, the association between AD and the other variants in the region was no longer significant ($p > 0.05$). *PSEN1* p.G206A was first identified in a few Caribbean Hispanic families,^{40,41} and a follow-up study identified *PSEN1* p.G206A in 70 families of Caribbean Hispanic ancestry.⁴² A more recent study of early-onset AD in Hispanics in Florida found that 13 out of 27 participants (48.1%) were *PSEN1* p.G206A carriers.⁴³ The allele frequency of this mutation is ultra-rare, only 1 in 1741 to 3790 individuals carries a *PSEN1* p.G206A in the Trans-Omics for Precision Medicine (TOPMed) Freeze 8⁴⁴ and Genome Aggregation Databases (gnomAD) v3.1.2³⁶ data, respectively. And while it is still ultra-rare, it is more common in Latino/Admixed Americans (1 in 423) within gnomAD v3.1.2. We observed 17 *PSEN1* p.G206A carriers in the entire ADSP R3 dataset (1 in 786), of which 16 individuals have AD, and all *PSEN1* p.G206A carriers are reported as Hispanic. Only one pair of *PSEN1* p.G206A carriers were inferred to be related (Figure S7 in supporting information). Among those with AD, we observed that *PSEN1* p.G206A carriers have an earlier age of diagnosis of AD (58.6 ± 7.6 years old) compared to non-carriers (74.7 ± 10.4 years old; $p = 1.1 \times 10^{-10}$).

Although previous studies have suggested that *PSEN1* p.G206A originated in Caribbean Hispanics,^{40,41} we do not have detailed data on the geographic location of most ADSP participants. We applied PCA to decipher the origin of the *PSEN1* p.G206A allele. PCA captures the human population structure associated with ancestry.⁴⁵ Therefore, we placed *PSEN1* p.G206A carriers into a global context using PCA of carriers and reference samples in the 1000 Genomes Project²⁵ from 26 populations, including 139 Puerto Ricans. Expectedly, the 17 *PSEN1* p.G206A carriers are closer to Puerto Rican reference samples in the American group according to the Euclidean distance of the first five PCs (Figure S8 in supporting information); individuals with at least 21 of 43 haplotype-associated SNVs are similarly placed near Puerto Ricans in PC space with greater dispersion (Figure S9 in supporting information). A global ancestry analysis revealed that all *PSEN1* p.G206A carriers have a higher proportion of European ($73.82\% \pm 6.85\%$) than African ancestry ($16.05\% \pm 5.72\%$). However, we observed that the chr14 risk haplotype including *PSEN1* p.G206A is inherited on an African-derived haplotype (Figure S10 in supporting information). This conclusion is based on results from RFMIX of combining our dataset with 2504 reference genomes from the 1000 Genomes Project. As *PSEN1* p.G206A is not detected in any individual of the AA subgroup while the other chr14 haplotype-defining variants have higher allele frequencies in the AA subgroup; *PSEN1* p.G206A may be too rare to be detected on AA subgroup or may have arisen in Puerto Rico⁴⁶ locally based on a founder event of an African haplotype.

Of note, one female *PSEN1* p.G206A carrier, whose *APOE* genotype is $\epsilon 3/\epsilon 3$ and global ancestry is 76.94% European genomes, was diagnosed as an AD patient at the age of 74 years, 15 years older than the average age at onset of AD among *PSEN1* p.G206A carriers. This raises the question of whether any other protective variants counteract the impact of mutations in the haplotype. To the best of our knowledge, this is the first report of a significant association between AD and this 3Mbp haplotype, which was linked to onset age 20 years younger.^{40,41}

3.4 | Association of aggregates of rare variants in genes

To determine whether aggregated of rare variants in genes associated with AD status, we performed gene-centric aggregates of coding variants for each protein-coding gene using five functional variant categories (pLoF, missense, disruptive missense, pLoF and disruptive missense, and synonymous; Figure S11 in supporting information). Unsurprisingly, we observed several significant gene-based associations ($p < 5 \times 10^{-7}$) in 14q24 in the HIS individuals (Table S12 in supporting information), which was also confirmed in the replication ($p < 0.15 \times 10^{-03}$, Table S5). Additionally, we observed that *PSEN1* was associated with AD ($p = 4.1 \times 10^{-8}$) in the pooled samples analysis when aggregating rare loss of function and disruptive missense variants (Figure 3). Several sensitivity analyses were performed in the pooled-population individuals. First, after excluding *PSEN1* p.G206A (rs63750082) from the gene-based tests, we observed a significant association between the aggregation of rare loss-of-function and disruptive missense variants in *PSEN1* and AD (OR 2.02, 95% confidence interval [CI] 1.3–3.0, $p = 7.8 \times 10^{-4}$). Second, we adjusted for *PSEN1* p.G206A, and observed almost identical results to the association without the adjustment of *PSEN1* p.G206A (OR 2.02, 95% CI 1.3–3.0, $p = 7.6 \times 10^{-4}$). Finally, none of the deleterious variants in *PSEN1* were in the list of significant chr14q24.2 variants reported in Table S11. These results suggest that there may be additional rare variants in *PSEN1* that contribute to the observed association with AD, particularly in the NHW subset.

We observed a few suggestive gene-based associations driven by rare coding variants ($5 \times 10^{-7} < p < 5 \times 10^{-5}$). Significant association signals between AD and *ABCA7* ($p = 5.4 \times 10^{-6}$) in the NHW analysis and *SPTLC2* ($p = 1.0 \times 10^{-5}$) in the HIS analysis were observed using multiple sets of rare coding variant aggregates (i.e., loss of function plus disruptive missense as well as missense, Table S12). There has been a growing interest in studying *ABCA7* due to accumulating in vitro and in vivo studies supporting the potential contribution to AD-related phenotypes.^{29,47} In the NHW subgroup, there are 273 rare exonic variants in *ABCA7* (including 186 non-synonymous, 9 stop-gain, 7 frameshift, and 3 non-frameshift; Table S13 in supporting information), and 90 of them are not singletons (including 67 non-synonymous, 2 stop-gain, and 4 frameshift; MAF: 0.000068–0.0075). Two frameshift deletions, rs547447016 (chr19:1047508:AGGAGCAG:A; $N_{\text{case}} = 30$ and $N_{\text{control}} = 9$) and rs538591288 (chr19:1055908:CT:C; $N_{\text{case}} = 20$ and $N_{\text{control}} = 8$), have

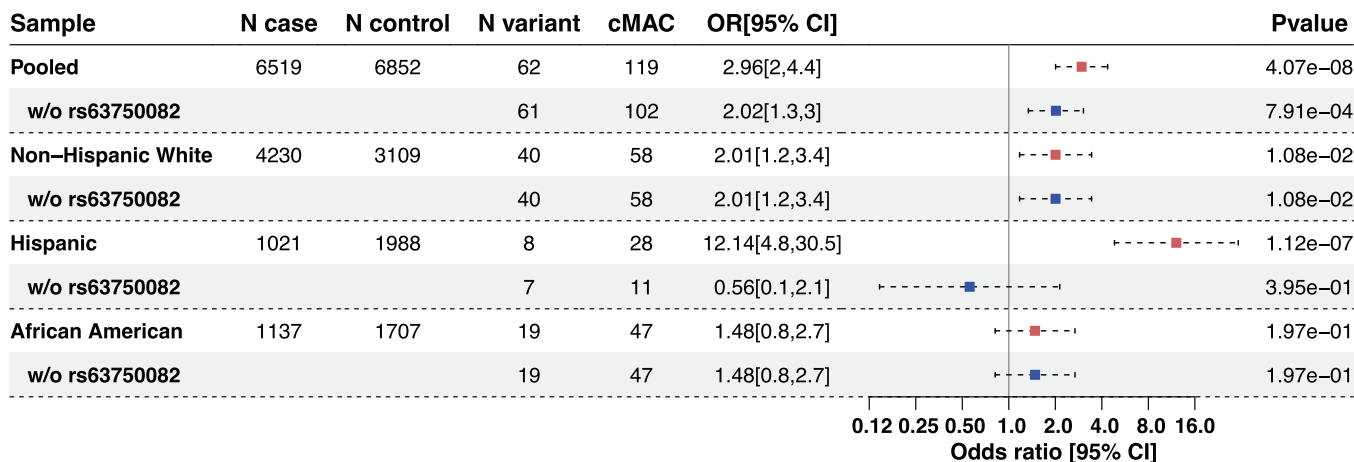


FIGURE 3 Association of *PSEN1* with AD by subgroup, with and without p.G206A (rs63750082) included. We performed aggregated-based association testing for variants within *PSEN1* in the pooled sample as well as in each subgroup. We then removed the p.G206A variant (rs63750082) from the aggregation-based test. We observed that the odds ratio and *p* value reduced significantly after excluding rs63750082 in the Hispanic subset but the results remained consistent in the NHW and AA subsets. AA, Black or African American; AD, Alzheimer's disease; CI, confidence interval; cMAC, cumulative minor allele counts across the variants contributing to the analysis; N variant, number of variants contributing to the analysis; NHW, non-Hispanic White.

been reported in previous studies^{48–50} and experimentally validated. Another two, rs779501556 (chr19:1046404:CGT:C; $N_{\text{case}} = 2$ and $N_{\text{control}} = 0$) and rs745871063 (chr19:1054250:AG:A; $N_{\text{case}} = 0$ and $N_{\text{control}} = 2$), are novel and ultra-rare. Nine of the stop-gain SNVs in our ABCA7 test (Table S13) were previously identified associated with AD^{48,51–55} or autism.⁵⁶

SPTLC2 (chr14:77,505,997–77,616,637) encodes a protein involved in the biosynthesis of sphingolipids, and there is some evidence to suggest that changes in sphingolipid metabolism may be associated with AD.^{57,58} Specifically, previous studies have suggested that alterations in the levels of specific sphingolipids may contribute to the development or progression of the disease.^{57,59} However, the role of *SPTLC2* in AD is not yet fully understood, and research in this area is ongoing. Our WGS study identified a suggestive significant association between the aggregate of rare disruptive missense and loss-of-function variants in *SPTLC2* (Table S14 in supporting information) with AD in the HIS subgroup ($p = 1.0 \times 10^{-5}$ and 1.0×10^{-5}), respectively.

3.5 | Association of aggregates of rare variants in non-coding sets

We next performed gene-centric aggregates of rare non-coding variants using eight functional variant categories (promoter or enhancer overlaid with CAGE or DHS sites, UTR, upstream, downstream, and non-coding RNA genes) to determine whether aggregated of rare non-coding variants associated with AD (Figure S11). We observed rare non-coding variant aggregates associated with AD near *TOMM40* ($p = 7.2 \times 10^{-8}$) and *PSEN1* ($p = 2.4 \times 10^{-11}$ to 3.2×10^{-8}) regions in the pooled samples and HIS individuals (Table 3). After conditioning on the number of *APOE* $\epsilon 2$ and $\epsilon 4$ alleles, there was an attenuation of the results for *TOMM40* in the pooled samples analysis, but the

association remained ($p_{\text{adj}} < 7.1 \times 10^{-6}$). In our replication data, we observed *TOMM40* associated with AD ($p = 2.8 \times 10^{-03}$). Compared to the pooled samples analysis, we observed that none of subgroup-specific associations reached a GW significant level for our rare variant aggregation tests ($p < 1 \times 10^{-7}$); however, we observed suggestive association signals in the AA and NHW individuals with variants in the promoter of *TOMM40*, $p = 2.8 \times 10^{-4}$ ($p_{\text{adj}} = 0.010$) and 4.4×10^{-3} ($p_{\text{adj}} = 0.018$), respectively, whereas there was not an association in HIS individuals ($p = 0.41$, $p_{\text{adj}} = 0.45$). After adjusting for *PSEN1* p.G206A, we observed that the association of rare non-coding variants in the promoter of *PSEN1* was no longer significant ($p > 0.05$) in pooled samples and HIS analyses. This suggests these rare non-coding variants are on the same haplotype as the rare coding variants in *PSEN1*; this was confirmed by local ancestry analysis (Figure S10).

4 | DISCUSSION

WGS data allow for the testing of both common and rare genetic variations that may be unique to individuals or populations and provide a powerful tool for identifying genetic variation that may be missed by traditional genotyping methods. Because the ADSP includes the largest sample of diverse participants with WGS and AD status, we designed our study to perform association testing across all participants and within subgroups defined by reported race/ethnicity. The pooled samples analysis is most powerful for detecting associations when there are similar effects across ancestries, while the subgroup-specific analyses are able to detect subgroup-specific effects. Through our analysis framework, we were able to adequately control for the diversity within ADSP and leveraged ADSP WGS to learn more about known loci for AD, including population-specific genetic signals.

TABLE 3 Aggregates of rare variants in non-coding sets with AD.

Group ^a	Gene name	Chr	Category	# Variants	STAAR-O p value ^a
Pooled samples	TOMM40	19	Promoter (DHS)	134	7.2×10^{-8}
Pooled samples	ELMSAN1	14	Enhancer (DHS)	1133	1.8×10^{-9}
Pooled samples	EIF2B2	14	Enhancer (DHS)	1240	3.2×10^{-8}
Pooled samples	MIR4505	14	ncRNA	7	2.4×10^{-11}
HIS	PTGR2	14	Promoter (CAGE and DHS)	7	8.85×10^{-12}
HIS	ELMSAN1	14	Enhancer (DHS)	366	3.1×10^{-11}
HIS	PTGR2	14	Enhancer (CAGE)	153	5.9×10^{-11}
HIS	ACOT6	14	Enhancer (DHS)	33	4.2×10^{-10}
HIS	ELMSAN1	14	Promoter (DHS)	55	8.8×10^{-10}
HIS	ACOT4	14	Promoter (CAGE)	6	9.3×10^{-10}
HIS	ACOT4	14	Enhancer (CAGE)	9	9.8×10^{-10}

Abbreviations: CAGE, Cap Analysis of Gene Expression; DHS, DNase Hypersensitivity; HIS, Hispanic.

^aWhere more than one rare non-coding aggregate was significant for a linked gene, the most significant p value is reported for each category type.

As anticipated, our single variant GWAS showed the strongest associations with AD in *APOE* across all groups as well as *BIN1* for the pooled samples and NHW. We observed that genetic variation in *LINC00320* was associated with AD in AA. We identified 44 genetic variants on 14q24 in HIS (MAF: 0.005–0.012), and aggregates of rare variants analysis in HIS and pooled samples confirmed the region, which includes p.G206A in *PSEN1*, a well-known early-onset AD mutation.^{40–43} Our PCA analysis indicates that *PSEN1* p.G206A carriers are closer to Puerto Ricans, consistent with previous studies,⁴⁶ while local ancestry analysis, however, pointed out that the local haplotype is derived from African genomes.

The analysis of coding rare variants identified suggestive associations in *ABCA7* in NHW and *SPTLC2* in HIS with AD. Rare coding variants in *ABCA7* have been found associated with AD risk.^{60–62} From our analysis, a deeper investigation of *ABCA7* revealed two frameshift deletions, rs547447016 and rs538591288, that were reported in previous studies^{48–50} and validated here, while rs779501556 and rs745871063 are variants newly associated with AD in the current analysis, indicating distinct *ABCA7* variants in multiple subgroups associated with AD risk. The non-coding rare variants in the promoter of *TOMM40* were identified as significant in the pooled samples analysis and confirmed to be distinct from the *APOE* haplotypes.

There are some limitations to our study. First, defining subgroups based on reported race and ethnicity, which was used in this study, may not be recommended as the best practice,⁶³ we observed a high consistency in reported and genetically defined subgroups, particularly for the AA and NHW subgroups (Table S3). Furthermore, humans cannot be grouped into discrete categories, and we acknowledge heterogeneity in our subgroup-specific analyses, and that the heterogeneity varies among the subgroups. We clustered samples using the Euclidean distance of PCA between each individual and the three 1000 Genomes

Project reference populations, EUR, AFR, and EAS (Figure S1). The PCA-based clusters are similar to the results using global ancestry analysis, 97.78% reported AA are inferred as having a majority of African ancestry, and 99.43 reported NHW are inferred as having a majority of European ancestry. As to the reported HIS, 74.74%, 20.51%, 2.43%, and 1.46% are inferred by GRAF-pop, a global ancestry inference, as having ancestries of Latin American 1, African American, Latin American 2, and European, respectively from GRAF-pop.²⁶ Association analyses on genetically defined subgroups gave similar results. Second, despite being the largest sample with WGS ascertained on AD status, we still have limited power to detect associations with AD, particularly with rare non-coding variants in which the aggregation unit is not as well defined as with rare coding variants. We acknowledge there are much larger sample sizes using GWAS for AD to assess the contribution of single variants.¹² In a prior study, we restricted to variants previously identified by GWAS loci and implicated genes using WGS data.⁶⁴ Third, each study contributing to the ADSP adjudicates their participants, which could lead to differences in the definition of AD across the contributing studies and a diminished power to detect associations with AD status.

In conclusion, we have comprehensively analyzed up to 13,371 diverse individuals with WGS for AD and observed common and rare variants associated with AD.

AUTHOR CONTRIBUTIONS

Wan-Ping Lee, Seung Hoan Choi, Margaret G. Shea, Beth A. Dombroski, Frederick Nussetor, and Gina M. Peloso performed statistical analyses. Nancy L. Heard-Costa, Amanda B. Kuzma, Lindsay A. Farrer, Margaret Pericak-Vance, Li-San Wang, Jonathan L. Haines, Gina M. Peloso, and Xiaoling Zhang performed phenotype acquisition and/or harmonization. Wan-Ping Lee, Achilleas N. Pitsillides, A.M.B., Yuk Yee Leung, Honghuang Lin, William S.

Bush, Eden Martin, Elizabeth E. Blue, Xiaoling Zhang, Margaret Pericak-Vance, Li-San Wang, Anita L. Destefano, Jonathan L. Haines, and Gina M. Peloso performed genotype acquisition and/or QC. Wan-Ping Lee, Seung Hoan Choi, Po-Liang Cheng, Hui Wang, Honghuang Lin, Josée Dupuis, Lindsay A. Farrer, Elizabeth E. Blue, Xiaoling Zhang, Myriam Fornage, Eden Martin, Ellen M. Wijsman, Li-San Wang, Anita L. Destefano, Jonathan L. Haines, and Gina M. Peloso interpreted results. Wan-Ping Lee and Gina M. Peloso wrote the first draft of the manuscript. All authors read, critically revised, and approved the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest. Author disclosures are available in the [supporting information](#).

DATA AVAILABILITY STATEMENT

ADSP R3 VCFs: <https://dss.niagads.org/datasets/ng00067/1000>
Genomes Project VCFs: http://ftp.1000genomes.ebi.ac.uk/vol1/ftp/data_collections/1000G_2504_high_coverage/working/20201028_3202_raw_GT_with_annot/

Full summary statistics will be made available through NIAGADS upon publication.

CONSENT STATEMENT

All human subjects included in this study provided informed consent.

CODE AVAILABILITY

https://github.com/wanpinglee/CADRE_CHARGE_ADSP17K

https://github.com/seuchoi/STAARpipeline_plugin

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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